EBOOK

Securing the Future: How Databricks Powers Financial Services



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Introduction



Security and compliance stakes in financial services have never been higher. According to recent industry surveys, **63%** of banking risk leaders now cite cyber risk as their top concern. Fraud has also reemerged as a major threat, ranking among the top five risks for **44%** of respondents. In the past 18 months alone, over **94%** of banks have experienced check fraud, and **61%** have encountered digital payments fraud, highlighting the urgent need for stronger defenses and faster response capabilities. As financial institutions face a growing web of regulatory, operational, and cybersecurity risks, they must protect sensitive data, maintain customer trust, and continue to innovate responsibly.

At Databricks, we know that data is one of your most valuable assets. Our product and security teams work together to deliver an enterprise-grade Data Intelligence Platform that enables you to defend against security risks and meet your compliance obligations. Over the past year, we've continued to expand our security and compliance offerings to meet the evolving needs of regulated industries, privately connect to external resources, support your zero trust initiatives, and manage AI risks.

This ebook explores how Databricks empowers financial services firms to secure their data, streamline compliance and confidently embrace the future of AI, turning security, compliance, governance and cybersecurity into a strategic advantage.

Security and Cybersecurity in the Databricks Data Intelligence Platform

Cybersecurity is foundational in the financial sector — critical for protecting sensitive data, preventing financial loss, maintaining consumer trust, and ensuring regulatory compliance in an industry that's a prime target for cybercriminals. It's no surprise that bank leaders identify it as a top risk, and cite fraud as a major concern.

Databricks's multi-layered, defense-in-depth approach ensures that your data and AI assets remain protected at every stage of their lifecycle:

- **Identity and access management**
  Single sign-on (SSO), multi-factor authentication (MFA), least privilege access and role-based controls.
- **Network security**
  Customer-managed VPCs, IP access lists, network isolation and firewall controls.
- **Data protection**
  Industry-standard encryption for data at rest and in transit, customer-managed keys and data masking.
- **Secret management**
  Secure storage and management of credentials and API keys.
- **Audit logging and monitoring**
  Comprehensive visibility into user activities, access patterns and system changes.
- **Real-time threat detection**
  Continuous monitoring, anomaly detection and alerting for suspicious activities.
- **Operationalized cybersecurity**
  Automated risk discovery, security testing and real-time enforcement across AI and MLOps pipelines, including controls for AI/ML-specific risks (e.g., model theft, prompt injection).
- **Disaster recovery and business continuity**
  Regular backups, recovery plans and redundant systems to minimize downtime and data loss.
- **Continuous innovation**
  Security capabilities evolve with the threat landscape, keeping you a step ahead of emerging risks and regulatory demands.

Compliance and Regulatory Support

Navigating the complex web of financial regulations is a constant challenge. Databricks simplifies compliance management:

- **Industry standards compliance**
  Built-in controls for NIST, ISO 42001, HITRUST, ENISA (EU cybersecurity), K-FSI (Korea Financial Security Institute), and more.
- **Audit logging and monitoring**
  Full traceability for regulatory audits and incident investigations.
- **Data lineage and traceability**
  Track the origin and transformation of data throughout its lifecycle, ensuring accountability and transparency.
- **Regulatory reporting support**
  Automate the creation of accurate, timely reports for regulatory bodies.



Databricks helps financial institutions address requirements from frameworks, such as GDPR (data privacy), DORA (digital operational resilience), PCI DSS (payment security), SOX (financial reporting compliance), FINRA (US securities regulation), AML (anti-money laundering compliance), and Basel III/IV (banking capital and liquidity requirements).

Governance with Unity Catalog

Effective data governance is crucial for maintaining trust and maximizing the value of your data assets. Unity Catalog provides:

- **Unified governance**
  A single source of truth for all governing data, AI and analytics across your organization.
- **Fine-grained access controls**
  Precise controls over who can access what data and at what time period.
- **Data discovery and metadata management**
  A way to easily find, understand and manage your data assets.
- **Identity provider integration**
  Compatibility with your existing identity management systems.



Best Practices for Risk Reduction and Compliance

To maximize security, reduce risk and optimize compliance, Databricks recommends the following best practices, which are aligned with industry standards and proven strategies.

- **Identity and access management**
  Enforce least privilege, use SSO and MFA, limit admin users.
- **Data protection**
  Encrypt data at rest and in transit, use customer-managed keys, centralize governance.
- **Network and endpoint security**
  Isolate sensitive workloads, restrict access, use firewalls.
- **Monitoring and threat detection**
  Enable verbose audit logging, integrate with SIEM tools, automate security analysis.
- **Governance and compliance**
  Assign Unity Catalog securables, implement fine-grained controls, test disaster recovery plans.
- **AI/ML-specific controls**
  Secure models, monitor usage, apply AI guardrails.



DASF in Action: Case Study

Coastal Community Bank leveraged DASF to systematically identify and address security, governance, and compliance risks as they rolled out AI initiatives.

COASTAL COMMUNITY BANK®

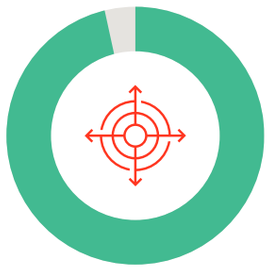
KEY OUTCOMES

- Board-driven AI governance and targeted risk controls
- Seamless standards alignment for easy compliance

BUSINESS RESULTS



66%
decrease in time to complete vendor due diligence reviews



94%
accuracy of customer complaints classifier with 100% coverage

An actionable framework for managing AI security

The Databricks AI Security Framework (DASF) 2.0, and the Databricks AI Governance Framework (DAGF v1.0) are purpose-built for modern finance. DASF 2.0 addresses 62 technical security risks across 12 core AI components, with controls mapped to global standards like MITRE, OWASP, NIST, ISO, and HITRUST. DAGF v1.0 adds robust AI governance with five pillars and 43 actionable considerations, helping organizations ensure transparency, accountability, and compliance in AI systems.

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Why Databricks for Financial Services

Databricks stands out for financial institutions with:



Purpose-built solutions

Tailored to address the unique challenges of the finance industry.



Unmatched scalability and performance

Process and easily analyze massive datasets, enabling real-time insights and decision-making.



Collaborative environment

Break down silos between data scientists, analysts and engineers, fostering innovation and efficiency.



Continuous innovation

Stay ahead of the curve with our commitment to developing cutting-edge features and capabilities.



Robust partner ecosystem

Integrate leading compliance, security and governance solutions for faster, more effective risk management.

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Future-Proofing Financial Services with Databricks

As the landscape evolves, Databricks helps you:

- Adapt to changing regulations with a flexible platform.
- Harness AI for enhanced security and faster fraud detection — AI-driven tools now reduce detection time and improve accuracy.
- Prepare for tomorrow's challenges with a robust, scalable data foundation.

At the 2025 Data + AI Summit, Databricks announced major security and compliance innovations:

- MLflow 3.0
  End-to-end model governance for trustworthy AI decisions.
- Mosaic AI Agent Bricks
  Automate compliance checks and fraud detection, reducing errors and freeing teams for strategic work.
- Security upgrades
  Serverless Egress Control, Private Link, and Multi-Key Protection for data security at scale.
- Unity Catalog with full Apache Iceberg™ support
  Unified governance across all data assets, simplifying audits and regulatory reporting.
- Lakebridge
  Free, AI-powered migration path for modernizing data warehouses, ensuring speed and security.

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Conclusion

In today's financial landscape, simply checking the box on regulatory requirements is no longer enough. Recent industry events have shown that risks often hide in plain sight, and that true resilience comes not from compliance for its own sake, but from ongoing vigilance, transparency and thoughtful risk management.

Financial institutions need partners who understand that robust security, proactive governance and strong cybersecurity are foundational to customer trust and long-term growth. Databricks delivers a comprehensive platform that helps organizations like yours anticipate and mitigate emerging risks, simplify compliance and embed a culture of accountability across data and AI initiatives.

By choosing Databricks, you gain:

- **Multi-layered security and governance for sensitive data and AI systems**
- **Simplified compliance workflows and reduced regulatory overhead**
- **Full visibility and auditability across data and AI systems**
- **The ability to innovate with confidence and respond quickly to new threats**

The future of finance will be led by institutions that balance innovation with rigorous governance and treat security not like a checkbox, but as a continuous, strategic commitment. Databricks helps financial companies build that future with our unified solution designed to keep your data and AI secure, compliant and adaptable to what's next.

Join the financial institutions redefining what responsible AI looks like. Together, we can create a resilient, transparent and future-ready financial ecosystem.

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Databricks is the data and AI company. More than 10,000 organizations worldwide — including Amgen, Providence, Humana and over 60% of the Fortune 500 — rely on the Databricks Data Intelligence Platform to unify and democratize data, analytics and AI. Databricks is headquartered in San Francisco, with offices around the globe, and was founded by the original creators of Lakehouse, Apache Spark™, Delta Lake and MLflow. To learn more, follow Databricks on X, LinkedIn and Facebook.



Get started with a free trial of Databricks and start building data applications today.

START YOUR FREE TRIAL

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